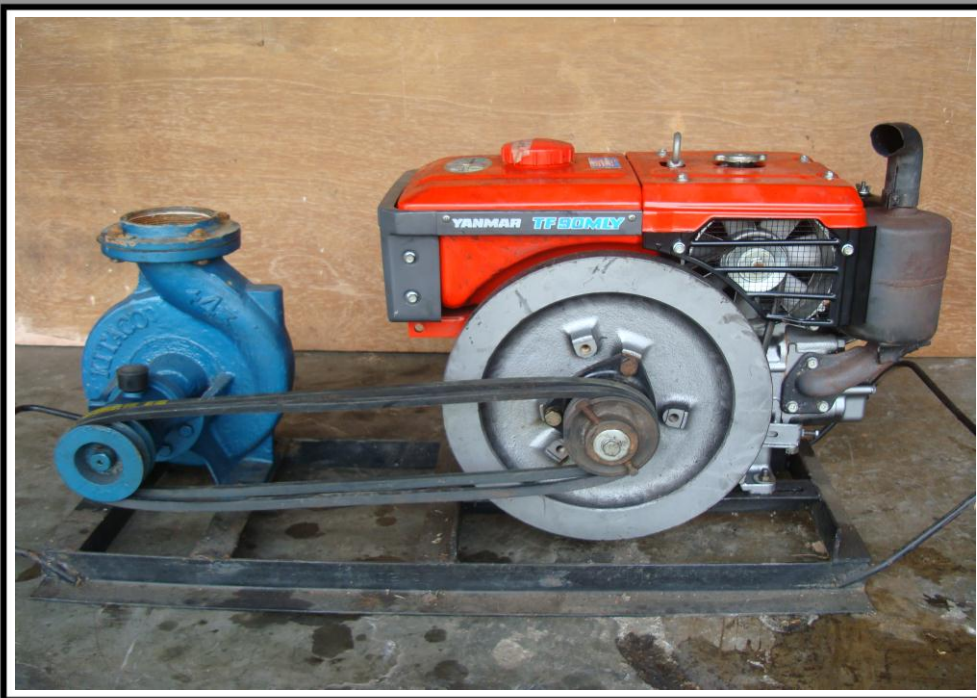




AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 16



**KITACO 4x4 MS coupled to
YANMAR TF90MLY Diesel Engine**

KITACO 4x4 MS Centrifugal Pump coupled to a YANMAR TF90MLY Diesel Engine

Test Report No.	:	2013 - 16
Date of Test	:	February 21, 2013
Test Valid Until	:	February 21, 2018
Brand	:	KITACO
Model	:	4x4 MS
Serial No.	:	No data available
Type and Size	:	Single-end suction, 100mm x 100mm, Centrifugal pump
Manufacturer	:	Made in China
Test Requested by	:	EQUITY MACHINERIES, INC. National Highway, Cauayan City, Isabela
Machine Classification	:	Commercial

SUMMARY

The KITACO 4x4 MS centrifugal pump coupled to YANMAR TF90MLY diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with actual diameters of 101.6 mm (4.0 inches) and two pieces of V-belts. The engine and pump were mounted on a steel base.

At zero discharge condition, the Maximum Total Head (H) was 32.5 m. The shaft speeds of the pump and engine at this point were 2485 and 2452 rpm, respectively, with Engine Fuel Consumption Rate of 1.46 L/h. The Maximum Discharge Capacity of 22.5 L/s occurred at 9.41m Total Head and shaft speeds of 2047 rpm and 2050 rpm for the pump and engine, respectively. The Fuel Consumption Rate at this point was 2.46 L/h.

The Maximum Fuel Consumption Rate of 2.46 L/h occurred at the point of Maximum Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.19 m. The Net Positive Suction Head Available at this point was 1.81 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pumpset performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. **Figure 1** shows the pump mounted on the laboratory test set-up.

The test was conducted on February 21, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. KITACO 4x4 MS Centrifugal pump coupled to YANMAR TF90MLY Diesel engine on the test set-up

METHOD OF TEST

The pump was tested following the “ Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The KITACO 4x4 MS Centrifugal pump coupled to YANMAR TF90MLY Diesel Engine shown in **Figure 2**, was a single-end suction, 100 mm x 100 mm, centrifugal pump coupled to a 7.10 kW YANMAR TF90MLY water-cooled diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine and pump pulleys were both 101.6 mm (4.0 inches). The suction inlet and discharge outlet were threaded flanges and designed for a 100 mm diameter pipe. The impeller was a closed-type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a mechanical seal. The pump shaft rotated on ball bearings kept in place by a bearing housing bolted to the casing.

The primemover, a YANMAR TF90MLY Diesel engine, was an 7.10 kW (9.50 hp), four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized oil bath type air cleaner. Lubrication system was force-feed type. Cooling was done through the circulation of water from the cylinder jacket to the radiator. It was started by a hand crank system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.

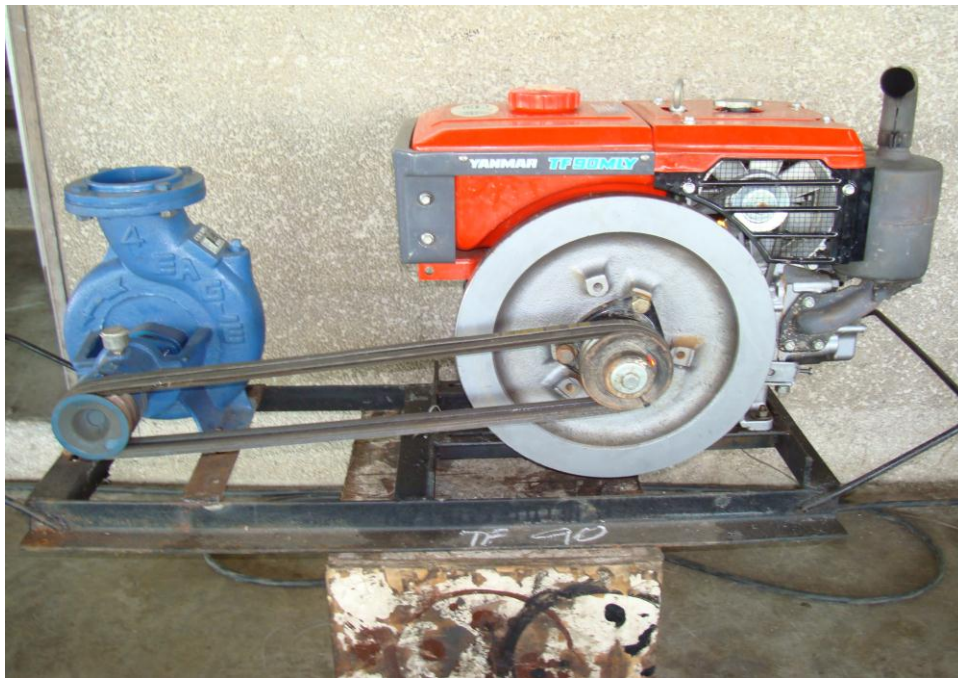


Figure 2. KITACO 4x4 MS Centrifugal Pump coupled to a YANMAR TF90MLY Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KITACO 4x4 MS Centrifugal Pump coupled to a YANMAR TF90MLY Diesel Engine

1.	Brand	:	KITACO
2.	Model	:	4 x 4 MS
3.	Serial No.	:	No data available
4.	Make	:	Made in China
5.	Type	:	Single-end suction, Centrifugal pump
6.	Size (mm)	:	100 x 100
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	1370
7.2	Width (mm)	:	460
7.3	Height (mm)	:	540
7.4	Weight (kg)	:	160.0
8.	Impeller		
8.1	Type	:	Closed
8.2	Diameter (mm)	:	200
8.3	Width (mm)	:	17
8.4	No. of vanes	:	9
8.5	Material	:	Brass
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	100
9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	100
10.3	Material	:	Cast Iron
11.	Casing		
11.1	Type	:	Volute
11.2	Material	:	Cast Iron
12.	Stuffing box	:	Mechanical seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	No data available
15.	Rated Maximum Discharge Capacity (L/s)	:	No data available
16.	Rated Maximum Total Head (m)	:	No data available
17.	Primemover		
17.1	Brand	:	YANMAR
17.2	Model	:	TF90MLY
17.3	Serial No.	:	DJ9760
17.4	Make	:	YANMAR CO., LTD. Indonesia
17.5	Type	:	Four stroke, water-cooled single horizontal cylinder diesel engine
17.6	Rated Output Power (kW)		
	Maximum	:	7.10 @ 2400 rpm
	Continuous	:	6.34 @ 2400 rpm

Table 1. (Cont'n)

17.8	Bore x Stroke	:	85 x 87
17.9	Displacement Volume (cc)	:	493
17.10	Cooling System	:	Water-cooled
17.11	Lubricating System	:	Force-feed type
17.12	Starting System	:	Manual cranking lever
17.13	Air Cleaner	:	Oil bath
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pump Tested : KITACO 4 x 4 MS coupled to
YANMAR TF90MLY Diesel engine

Date of Test : February 21, 2013

Test Condition

Ambient Temperature (°C) : 26.7

Atmospheric Pressure (mb) : 1013

Pumping Water Temperature (°C) : 22.7

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP- TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	SYSTEM EFFICIENCY (%)
2452	2485	1.46	32.5	0	0	0
2350	2355	2.20	29.2	7.79	2.23	9.83
2279	2287	2.19	27.3	8.62	2.31	10.2
2254	2233	2.17	25.3	9.57	2.38	10.6
2150	2156	2.12	22.6	11.3	2.51	11.5
2118	2119	2.14	20.7	12.6	2.56	11.6
2140	2093	2.26	18.7	14.4	2.63	11.4
2055	2060	2.32	16.3	16.6	2.66	11.2
2032	2026	2.44	13.3	19.0	2.47	9.92
2050	2047	2.46	9.41	22.5	2.08	8.31

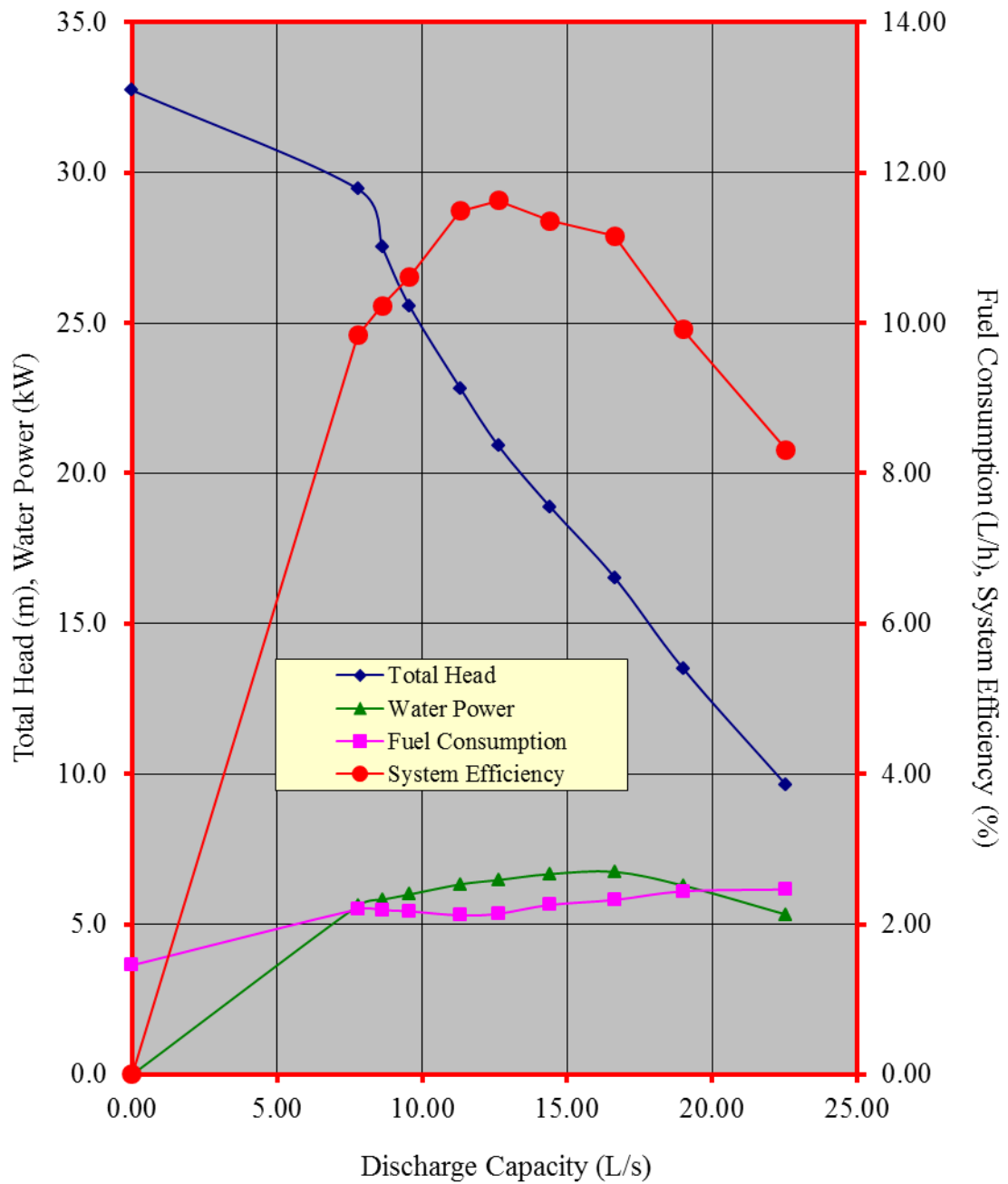


Figure 3. Performance characteristic curves of the KITACO 4 x 4MS centrifugal pump coupled to YANMAR TF90MLY Diesel Engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pump Tested : KITACO 4 x 4 MS Pump coupled to YANMAR TF90MLY Diesel engine

Date of Test : February 21, 2013

Test Condition

Ambient Temperature (°C) : 27.1

Atmospheric Pressure (mb) : 1013

Pumping Water Temperature (°C) : 25.0

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2049	2053	1.47	8.53	10.3	33.7
2163	2033	2.50	7.51	10.9	32.2
2125	2217	3.91	6.10	12.1	31.2
2063	2055	5.41	4.60	13.0	28.1
2313	2316	6.39	3.61	12.4	20.0
2482	2495	7.26	2.74	12.1	12.2
2500	2532	8.19	1.81	12.5	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power , IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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